

Chapter 3

Quadratic Variation and the Basic Itô Formula

Let B be a Brownian motion on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Our goal is to define an integral of the form

$$\int_0^T B_t(\omega) dB_t(\omega).$$

Since $t \mapsto B_t(\omega)$ is a uniformly continuous function on $[0, T]$, the naive approach would be to take a partition $0 = t_0 < t_1, \dots, < t_n = T$ of $[0, T]$ and to discretise the sample paths. Both

$$\phi_1(t, \omega) = \sum_j B_{t_j}(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

and

$$\phi_2(t, \omega) = \sum_j B_{t_{j+1}}(\omega) \mathbf{1}_{(t_j, t_{j+1}]}(t)$$

“reasonably” approximate the sample path $t \rightarrow B_t(\omega)$.¹ Since ϕ_1 and ϕ_2 are piecewise constant the integrals

$$\int_0^T \phi_i(t, \omega) dB_t(\omega) \quad (i = 1, 2)$$

make sense if we define $\int_s^u dB_t = B_u - B_s$. However,

$$\mathbb{E} \left[\int_0^T \phi_1(t, \omega) dB_t(\omega) \right] = \sum_j \mathbb{E} [B_{t_j} (B_{t_{j+1}} - B_{t_j})] = 0$$

¹We shall see later that only the first approximation is indeed “reasonable”.

while, using the martingale property of Brownian motion ($\mathbb{E}[B_{t_{j+1}}|\mathcal{F}_{t_j}] = B_{t_j}$)

$$\begin{aligned}
\mathbb{E}\left[\int_0^T \phi_2(t, \omega) dB_t(\omega)\right] &= \sum_j \mathbb{E}[B_{t_{j+1}}(B_{t_{j+1}} - B_{t_j})] \\
&= \sum_j \mathbb{E}[(B_{t_{j+1}} - B_{t_j})^2 + B_{t_{j+1}}B_{t_j} - B_{t_j}^2] \\
&= \sum_j \mathbb{E}[(B_{t_{j+1}} - B_{t_j})^2] + \sum_j [\mathbb{E}[B_{t_j}\mathbb{E}[B_{t_{j+1}}|\mathcal{F}_{t_j}]] - \mathbb{E}B_{t_j}^2] \\
&= T.
\end{aligned}$$

This suggests that we cannot define the above integral in the usual sense. In what follows we explore the reasons why this naive construction fails.

3.1 Quadratic variation and the Stieltjes Integral

Let us fix a *continuous* function $X : [0, T] \rightarrow \mathbb{R}^n$. We denote the components of X by X^i ($i = 1, \dots, n$). Our goal is to clarify under which conditions we can define an integral with respect to X analogous to the usual Riemann integral. To this end, we first introduce the concept of quadratic variation. Subsequently we introduce the Stieltjes integral.

Definition 3.1.1 A partition Π of the time interval $[0, T]$ is a set of points $0 = t_0 < t_1 < \dots < t_n = T$. The mesh is

$$|\Pi| := \max_i |t_i - t_{i-1}|.$$

Definition 3.1.2 • The (total) variation of the function $X^i : [0, T] \rightarrow \mathbb{R}$ along the partition $\Pi = \{t_0, \dots, t_n\}$ of $[0, T]$ is defined by

$$V_T(X^i, \Pi) := \sum_{i=1}^n |X^i(t_j) - X^i(t_{j-1})|$$

The (total) variation on $[0, T]$ is defined by

$$V_T(X^i) := \sup \{V_T(X^i, \Pi) : \Pi \text{ is a partition of } [0, T]\}.$$

If $V_T(X^i) < \infty$, then X^i is of bounded/finite variation on $[0, T]$.

- The co-variation process of the functions $X^j, X^k : [0, T] \rightarrow \mathbb{R}$ along the partition $\Pi = \{t_0, \dots, t_n\}$ of $[0, T]$ is given by

$$V_t^2(X^j, X^k, \Pi) = \sum_{t_{i-1}, t_i \in \Pi, t_i \leq t} (X^j(t_i) - X^j(t_{i-1}))(X^k(t_i) - X^k(t_{i-1})).$$

The quadratic variation process of X^i along Π is given by

$$V_t^2(X^i, \Pi) = V_t^2(X^i, X^i, \Pi), \quad t \in [0, T].$$

- We say that X is of continuous quadratic variation on $[0, T]$ if the limit

$$\langle X^j, X^k \rangle_t = \lim_{n \rightarrow \infty} V_t^2(X^j, X^k, \Pi^n)$$

exists for all $j, k = 1, \dots, n$ along a sequence of partitions $\{\Pi^n\}$ with $|\Pi^n| \rightarrow 0$ as $n \rightarrow \infty$ and the map $t \mapsto \langle X, Y \rangle_t$ is continuous. The quadratic variation process $\langle X^i \rangle$ of the function X^i is defined by

$$\langle X^i \rangle := \langle X^i, X^i \rangle.$$

Notice that we require the quadratic variation to exist only along a given sequence of partition. For the special case of Brownian sample paths we shall show that the choice of partition is irrelevant as long as the mesh sizes converge to zero sufficiently fast. In general, this need not be the case, though.

Remark 3.1.3 *Strictly speaking, we should have defined our co-variation processes along a partition Π as*

$$V_t^2(X^j, X^k, \Pi) = \sum_{t_{i-1}, t_i \in \Pi} (X^j(t_i \wedge t) - X^j(t_{i-1} \wedge t))(X^k(t_i \wedge t) - X^k(t_{i-1} \wedge t)).$$

In other words, when computing quantities like $V_t^2(X^j, X^k, \Pi)$ we assume that $t \in \Pi$. Since we only consider continuous processes, a simple 3ε -argument shows that the two definitions result in the same limit if $|\Pi| \rightarrow 0$. Our slightly sloppy definition saves us a lot of notation. In the sequel we follow the convention that $t \in \Pi$ when computing (co-)variations.

Every monotone real valued function defined on a bounded interval is of bounded variation, and every real-valued function of bounded variation defined on a bounded interval is bounded. The quadratic variation process $\langle X \rangle$ of a continuous real valued function X is increasing and satisfies $\langle X \rangle_0 = 0$. If $X \in \mathcal{C}^1$ with $X_0 = 0$, then

$$X_t = \int_0^t X'_s ds \quad \text{and} \quad V_t(X) = \int_0^t |X'_s| ds.$$

The next proposition shows that X has quadratic variation zero if it is of bounded variation. We deduce that all monotone processes are of trivial quadratic variation.

Proposition 3.1.4 *If $X : [0, T] \rightarrow \mathbb{R}$ is continuous and of bounded variation, then its quadratic variation process is trivial, i.e., identically equal to zero*

PROOF: For any partition $\Pi^n = (t_0^n, \dots, t_{m_n}^n)$ we have that

$$\begin{aligned} \sum_{t_{i-1}^n, t_i^n \in \Pi^n, t_i^n \leq t} (X(t_i^n) - X(t_{i-1}^n))^2 &\leq \sup_i |X(t_i^n) - X(t_{i-1}^n)| \cdot \sum_{t_{i-1}^n, t_i^n \in \Pi^n, t_i^n \leq t} |X(t_i^n) - X(t_{i-1}^n)| \\ &\leq \sup_i |X(t_i^n) - X(t_{i-1}^n)| V_T(X). \end{aligned}$$

Since X is continuous and hence uniformly continuous on compact sets

$$\lim_{n \rightarrow \infty} \sup_i |X(t_i^n) - X(t_{i-1}^n)| = 0 \quad \text{if} \quad |\Pi^n| \rightarrow 0.$$

□

The following properties are easily verified.

Exercise 25 *Let $X, Y : [0, T] \rightarrow \mathbb{R}$ be continuous and of continuous quadratic variation with respect to a common sequence of partitions $\{\Pi_n\}$. Then the following holds:*

- (i) The quadratic variation $\langle X, Y \rangle$ exists and is continuous (along $\{\Pi_n\}$) if and only if $\langle X + Y \rangle$ exists and is continuous.
- (ii) $\langle \cdot, \cdot \rangle$ defines a (not necessarily positive semi-definite) bilinear form along $\{\Pi_n\}$.
- (iii) The polarization identity holds along $\{\Pi_n\}$:

$$\langle X, Y \rangle = \frac{1}{2} (\langle X + Y \rangle - \langle X \rangle - \langle Y \rangle).$$

In particular, co-variation processes are differences of two non-decreasing processes and hence of bounded variation (cf. Exercise 26).

- (iv) The Cauchy-Schwarz inequality holds along $\{\Pi_n\}$:

$$|\langle X, Y \rangle| \leq \sqrt{\langle X \rangle \langle Y \rangle}.$$

The polarization identity along with the Cauchy-Schwarz inequality yields the following corollary.

Corollary 3.1.5 *Let X be continuous with continuous variation and A be of bounded variation. Then*

$$\langle X + A \rangle = \langle X \rangle.$$

PROOF: Since A is of finite variation $\langle A \rangle = 0$ Cauchy-Schwarz yields $|\langle X, A \rangle| = 0$. Hence $\langle X + A \rangle = \langle X \rangle$, due to the polarization identity. $\square$

Exercise 26 *Prove that a real-valued function is of bounded variation if and only if it can be represented as the difference of two non-decreasing functions. In particular, (co-)variation processes are of bounded variation.*

Bounded variation functions can be linked to measure theory and hence integrals w.r.t. bounded variation functions are well defined. To see this, we recall basic facts from measure theory (Analysis III). Let $\mathcal{X}$ be a set and $\mathcal{A}$ be a σ -algebra in $\mathcal{X}$, and let $\mu : \mathcal{A} \rightarrow \mathbb{R}$ be a measure on $\mathcal{A}$, i.e. a countably additive function. The measure can be decomposed as $\mu = \mu^+ - \mu^-$ (Hahn-Jordan decomposition) where the measures $\mu^\pm$ are concentrated on disjoint sets $\mathcal{X}^\pm$ of $\mathcal{X}$ with

$$\mu^+(X) \geq 0, \quad \mu^-(X) \leq 0 \quad \text{for any } X \in \mathcal{X}^\pm$$

and at least one of the measures is finite. The *total variation* of μ is given by $|\mu| := \mu^+ + \mu^-$ and the *total variation norm* is given by $\|\mu\| := |\mu|(\mathcal{X})$.

A measure on the Borel-field $\mathcal{B}(\mathcal{X})$ is called a Borel measure. For a *bounded* Borel measure $\mu : \mathcal{B} \rightarrow \mathbb{R}$ on the real line we can define the distribution function $F_\mu(t) := \mu((-\infty, t))$. If the measure is non-negative, then F_μ is increasing; else it follows from the Hahn-Jordan decomposition that F_μ can be represented as the difference of two increasing functions. Since μ is bounded, it follows that F_μ is of bounded variation. Conversely, it follows from the preceding exercise that we can associated with every function $X : [0, T] \rightarrow \mathbb{R}$ of bounded variation a bounded *signed* measure $\mu : \mathcal{B}_T \rightarrow \mathbb{R}$. Indeed, since X is of bounded variation, $X = X^+ - X^-$ where $X^\pm$ are non-decreasing functions. We can associated a non-negative measure $\mu^\pm : \mathcal{B}_T \rightarrow \mathbb{R}_+$ with $X^\pm$ via $\mu^\pm([a, b)) := X^\pm(b) - X^\pm(a)$ and so $\mu := \mu^+ - \mu^-$ is a signed measure.

Remark 3.1.6 *In view of the above, we can define integrals of (e.g. bounded or continuous) functions with respect to functions of bounded variation. In particular, the integral $\int_0^T f(t)d\langle X \rangle_t$ is well defined for any continuous $f : [0, T] \rightarrow \mathbb{R}$ if X has continuous quadratic variation.*

Another approach to define this integral is the Riemann-Stieltjes integral, defined in terms of Riemann-Stieltjes sums. To this end, let

$$X, Y : [0, T] \rightarrow \mathbb{R}$$

be real-valued functions and $\Pi = \{t_0 < t_1 < \dots < t_n\}$ be a partition of $[0, T]$. Let $\xi = (\xi_1, \dots, \xi_n)$ be a vector of points $\xi_k \in [t_{k-1}, t_k]$. We call

$$S(X, Y, \Pi, \xi) := \sum_{k=1}^n Y(\xi_k) (X(t_k) - X(t_{k-1}))$$

a *Riemann-Stieltjes (RS) sum*. A sequence $\{S(X, Y, \Pi^n, \xi^n)\}_{n \in \mathbb{N}}$ is called a *RS sequence* if $|\Pi^n| \rightarrow 0$ as $n \rightarrow \infty$. If the sequence converges to some (and hence unique - the argument is the same as for Riemann sums) limit, then the limit is called the *Riemann-Stieltjes integral of Y with respect to X* . The limit is denoted

$$\int_0^T Y(s) dX(s).$$

The limit satisfies the same properties (linearity, ...) as the Riemann integral. Moreover, if Y is bounded, then $|\int Y dX| \leq \|Y\|_\infty V_T(X)$. The following lemma is easily established.

Lemma 3.1.7 *Let $X_1, X_2, Y : [0, T] \rightarrow \mathbb{R}$ be real-valued functions. If the integrals $\int Y dX_i$ exists for $i = 1, 2$, then the integral $\int Y d(X_1 + X_2)$ exist and*

$$\int Y d(X_1 + X_2) = \int Y dX_1 + \int Y dX_2.$$

The RS integral exists whenever Y is continuous and X is of bounded variation as shown by the following theorem.

Theorem 3.1.8 *Let $Y : [0, T] \rightarrow \mathbb{R}$ be continuous and $X : [0, T] \rightarrow \mathbb{R}$ be of bounded variation. Then the RS integral $\int Y dX$ exists.*

PROOF: Since X is of bounded variation, it can be decomposed into the difference of two non-decreasing functions: $X = X_+ - X_-$ where $X_\pm : [0, T] \rightarrow \mathbb{R}$ is non-decreasing. The existence of the integrals $\int Y dX_\pm$ can be established in exactly the same manner as the existence of the Riemann integral. The existence of the integral $\int Y dX$ then follows from the preceding lemma. $\square$

The analysis of this section suggests that a Brownian sample path is typically not of bounded variation. We analyse the quadratic variation of Brownian motion in the following section.

3.2 Quadratic Variation of Brownian motion

We are now going to compute the quadratic variation of Brownian motion. It turns out that the quadratic variation can be computed as a *deterministic* limit for almost every Brownian sample path. This will allow us to compute integrals with respect to Brownian motion for a class of integrands that

are smooth functions of the Brownian motion using *deterministic* arguments. In a later chapter we extend the definition of the integral to a larger class of integrands using *probabilistic* arguments.

Theorem 3.2.1 *Let $\{\Pi^n\}$ be a sequence of partitions of $[0, T]$ such that $\sum_n |\Pi^n| < \infty$. Then*

$$\lim_{n \rightarrow \infty} V_t^2(B(\omega), \Pi^n) = t, \quad \text{for all } t \in [0, T] \quad \mathbb{P}\text{-a.s.}$$

PROOF: In proving the assertion we shall assume that t belongs to the partitions Π^n . The arguments can be made precise using a standard ε -argument. In a first step we prove L^2 -convergence. To this end, observe that independence of the increments along with $\mathbb{E}[(B_{t_i} - B_{t_{i-1}})^2] = t_i - t_{i-1}$ yields

$$\begin{aligned} \mathbb{E} \left[\sum_{t_i \in \Pi^n} (B_{t_i} - B_{t_{i-1}})^2 - t \right]^2 &= \mathbb{E} \left[\sum_{t_i \in \Pi^n} \{ (B_{t_i} - B_{t_{i-1}})^2 - (t_i - t_{i-1}) \} \right]^2 \\ &= \sum_{t_i \in \Pi^n} \mathbb{E} [(B_{t_i} - B_{t_{i-1}})^2 - (t_i - t_{i-1})]^2 \\ &= \sum_{t_i \in \Pi^n} \text{Var}((B_{t_i} - B_{t_{i-1}})^2). \end{aligned}$$

Since

$$B_{t_i} - B_{t_{i-1}} \sim N(0, t_i - t_{i-1}) \quad \text{it follows that} \quad \text{Var}((B_{t_i} - B_{t_{i-1}})^2) = 2(t_i - t_{i-1})^2.$$

Hence

$$\begin{aligned} \mathbb{E} \left[\sum_{t_i \in \Pi^n} (B_{t_i} - B_{t_{i-1}})^2 - t \right]^2 &= 2 \sum_{t_i \in \Pi^n} (t_i - t_{i-1})^2 \\ &\leq 2 \cdot |\Pi^n| \cdot t \end{aligned}$$

which converges to zero as $n \rightarrow \infty$. In particular, the quadratic variation $\langle B(\omega) \rangle_t$ exists for every $t \in [0, T]$ as an L^2 limit. This convergence holds without any additional assumptions on the partitions.

L^2 -convergence implies almost sure convergence along a subsequence but we actually need to show convergence of the entire sequence. For this we write

$$\mathbb{E}[V_t^2(B, \Pi^n)] = \mathbb{E} \left[\sum_{t_i \in \Pi^n} (B_{t_i} - B_{t_{i-1}})^2 \right] = \sum_{t_i \in \Pi^n} \mathbb{E}[(B_{t_i} - B_{t_{i-1}})^2] = t.$$

Thus, by Tchebysheff's inequality we have for every $\varepsilon > 0$ that

$$\mathbb{P}[|V_t^2(B, \Pi^n) - t| > \varepsilon] \leq \frac{\text{Var}(V_t^2(B, \Pi^n))}{\varepsilon^2}.$$

Since a Brownian sample path has independent increments

$$\text{Var}(V_t^2(B, \Pi^n)) = \sum_{t_i \in \Pi^n} \text{Var}((B_{t_i} - B_{t_{i-1}})^2) = 2 \sum_{t_i \in \Pi^n} (t_i - t_{i-1})^2 \leq 2T|\Pi^n|$$

so

$$\mathbb{P}[|V_t^2(B, \Pi^n) - t| > \varepsilon] \leq \frac{2T}{\varepsilon^2} |\Pi^n|.$$

Since $\sum_n |\Pi^n| < \infty$ it follows from the lemma by Borel and Cantelli that

$$\mathbb{P}[|V_t^2(B, \Pi^n) - t| > \varepsilon \text{ infinitely often}] = 0.$$

This shows that, for every $t \in [0, T]$ there exists a set N_t of measure zero such that

$$\lim_{n \rightarrow \infty} V_t^2(B(\omega), \Pi^n) = \langle B(\omega) \rangle_t = t \quad \text{for all } \omega \notin N_t.$$

Let us now put $\mathcal{N} := \bigcup_{t \in [0, T] \cap \mathbb{Q}} N_t$; this set has measure zero and

$$\lim_{n \rightarrow \infty} V_t^2(B(\omega), \Pi^n) = \langle B(\omega) \rangle_t = t \quad \text{for all } t \in [0, T] \cap \mathbb{Q}; \quad \omega \notin \mathcal{N}.$$

For any $t \notin [0, T] \cap \mathbb{Q}$ there are sequences $\{t_n\}, \{s_n\} \in [0, T] \cap \mathbb{Q}$ such that $t_n \uparrow t$ and $s_n \downarrow t$. Moreover, there exists a random variable n^* such that for any $\omega \notin \mathcal{N}$ and all $n \geq n^*(\omega)$,

$$t_n - \varepsilon \leq V_{t_n}^2(B(\omega), \Pi^n) \leq V_t^2(B(\omega), \Pi^n) \leq V_{s_n}^2(B(\omega), \Pi^n) \leq s_n + \varepsilon.$$

This proves the result. $\square$

Remark 3.2.2 *We emphasize that we used the same sequence of partition of every ω when computing the quadratic variation of Brownian sample paths. If we were allowed to choose different partitions for different sample paths, then we could obtain any quadratic variation, including the zero process. For instance, one can prove that for any continuous function $f : [0, 1] \rightarrow \mathbb{R}$ there exists a sequence of partitions $\{\Pi_n\}$ with mesh sizes going to zero, such that $V_t^2(f, \Pi_n) \rightarrow 0$ for all $t \in [0, 1]$.*

3.2.1 The basic Itô formula

For a given partition Π of the interval $[0, T]$ and a given continuous function $X : [0, T] \rightarrow \mathbb{R}^n$ of continuous quadratic variation we denote by

$$\Delta_i X^j := X_{t_{i+1}}^j - X_{t_i}^j$$

the increments of X^j along Π . If X were of bounded variation, then for every continuously differentiable function f on $\mathbb{R}^n$ we would have

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s.$$

If $\langle X \rangle \neq 0$, i.e., X is of unbounded variation, then we need a generalization that accounts for the “many small fluctuations generating the quadratic variation”. The Itô formula provides such a generalization. For its proof we first state without proof the following result on the weak convergence of signed measures²

Proposition 3.2.3 *Let $\{\mu_n\}$ be a sequence of signed measures on $[a, b]$. The sequence converges weakly to a signed measure μ if and only if $\sup_n \|\mu_n\| < \infty$ and every subsequence of the sequence of distribution functions $\{F_{\mu_n}\}$ contain a further subsequence that converges to F_μ at all points excepting points of an at most countable set.*

Corollary 3.2.4 *Let $\{\Pi^n\}$ be a sequence of partitions of $[0, T]$ with $|\Pi^n| \rightarrow 0$ and $g : [0, T] \rightarrow \mathbb{R}$ continuous. Then*

$$\lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n, t_i \leq t} g(t_i) \Delta_i X^j \Delta_i X^k = \int_0^t g(s) d\langle X^j, X^k \rangle_s \quad \text{for all } t \in [0, \infty).$$

²For the special case of probability measures the theorem simplifies to the well known Portemonteau theorem that states that a sequence of probability measures $\{\mu_n\}$ converges weakly to a probability measure μ if and only if the associated sequence of distribution functions $\{F_{\mu_n}\}$ converges to F_μ at any point of continuity of F_μ .

PROOF: We can define signed measures μ_n and μ on the real line by

$$\mu_n(dt) := \sum_{t_i \in \Pi^n} \delta_{t_i}(dt) \Delta_i X^j \Delta_i X^k \quad \text{and} \quad \mu(dt) := d\langle X^j, X^k \rangle_t.$$

The associated distribution functions are

$$\mu_n([0, t)) = \sum_{t_i \leq t} \Delta_i X^j \Delta_i X^k \quad \text{and} \quad \mu([0, t)) = \langle X^j, X^k \rangle_t$$

where the second equation follows from the continuity of the co-variation process. By assumption $\mu_n([0, t)) \rightarrow \mu([0, t))$ for every $t \in [0, T]$. Hence, the assertion follows from the above proposition along with the continuity of g and the fact that

$$\sum_{t_i \leq t} g(t_i) \Delta_i X^j \Delta_i X^k = \int_0^t g(s) \mu_n(ds) \quad \text{and} \quad \int_0^t g(s) d\langle X^j, X^k \rangle_s = \int_0^t g(s) \mu(ds).$$

□

Remark 3.2.5 *There is another way to prove Corollary 3.2.4 that does not require Proposition 3.2.3. The proof of Exercise 25(i) shows that we can decompose the measure μ_n as $\mu_n = \mu_n^+ - \mu_n^-$ where*

$$\mu_n^+(dt) := \sum_{t_i \in \Pi^n} \delta_{t_i}(dt) (\Delta_i(X^j + X^k))^2 \quad \text{and} \quad \mu_n^-(dt) := \sum_{t_i \in \Pi^n} \delta_{t_i}(dt) ((\Delta_i X^j)^2 + (\Delta_i X^k)^2)$$

are non-negative measures and hence probability measures after suitable scaling. Again by Exercise 25(i) we know that all the distribution functions converges pointwise. Convergence in $t = T$ shows that the total masses converge. Hence, the assertion follows from the Portemonteu theorem applied to the scaled measures. This argument works because we know that not only that the distribution function of μ_n but also the distribution functions associated with $\mu_n^\pm$ converge.

In what follows $C^{1,2}(\mathbb{R}^m \times \mathbb{R}^n)$ denotes the class of all continuous functions $F : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}$ whose first the partial derivatives on $\mathbb{R}^m$ and whose first and second partial derivatives exist on $\mathbb{R}^n$ exist and are continuous. The notation $F \in C^{1,2}([0, T] \times \mathbb{R}^n)$ indicates that $F \in C^{1,2}((0, T) \times \mathbb{R}^n)$ and that all partial derivatives extend continuously to $[0, T] \times \mathbb{R}^n$. We drop the domain of the functions if there is no risk of confusion.

We are now ready to state and prove a first version of Itô's formula.

Theorem 3.2.6 *(Basic Itô formula) Let $X : [0, T] \rightarrow \mathbb{R}^n$ be continuous, of continuous QV, and let $A : [0, T] \rightarrow \mathbb{R}^m$ be continuous and of bounded variation. Let $F : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}$ be of class $C^{1,2}$. Then*

$$\begin{aligned} F(A_t, X_t) &= F(A_0, X_0) + \int_0^t \nabla_a F(A_s, X_s) dA_s + \int_0^t \nabla_x F(A_s, X_s) dX_s \\ &\quad + \frac{1}{2} \sum_{j,k=1}^n \int_0^t \frac{\partial^2}{\partial x^j \partial x^k} F(A_s, X_s) d\langle X^j, X^k \rangle_s \end{aligned} \tag{3.1}$$

with

$$\int_0^t \nabla_x F(A_s, X_s) dX_s := \lim_n \sum_{t_i} \nabla_x F(A_{t_i}, X_{t_i}) \Delta_i X.$$

PROOF: We prove the theorem for $n = m = 1$; the general case follows from straightforward modifications. Let us put $F_a := \nabla_a F$, $F_x := \nabla_x F$ and $F_{xx} := \frac{\partial^2 F}{\partial x^2}$ and let $\{\Pi_n\}$ be a sequence of partitions of the interval $[0, T]$ whose mesh sizes tend to zero. Assuming for simplicity that $t \in \Pi^n$ we can write

$$\begin{aligned} F(A_t, X_t) - F(A_0, X_0) &= \sum_{t_i \in \Pi^n} \{F(A_{t_{i+1}}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_i})\} \\ &= \sum_{t_i \in \Pi^n} \{F(A_{t_{i+1}}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_{i+1}})\} \\ &\quad + \sum_{t_i \in \Pi^n} \{F(A_{t_i}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_i})\}. \end{aligned}$$

We study the two sums separately.

- As for the first sum we notice that because F is continuously differentiable in its first argument the mean value theorem yields $\tilde{a}_i \in (A_{t_i}, A_{t_{i+1}})$ ³ such that

$$F(A_{t_{i+1}}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_{i+1}}) = F_a(\tilde{a}_i, X_{t_{i+1}}) \Delta_i A.$$

Since F_a , A and X are continuous, so is the function $g(t) := F_a(A_t, X_t)$, and the definition of the Riemann-Stieltjes integral yields

$$\lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n} g(\xi_i) \Delta_i A = \int_0^t g(s) dA_s$$

where $t_i \leq \xi_i \leq t_{i+1}$. Furthermore, by continuity of A , if $|\Pi_n|$ is small enough, then

$$\max_i |g(t_{i+1}) - F_a(\tilde{a}_i, X_{t_{i+1}})| < \varepsilon.$$

This yields that

$$\begin{aligned} \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n} \{F(A_{t_{i+1}}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_{i+1}})\} &= \lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n} F_a(\tilde{a}_i, X_{t_{i+1}}) \Delta_i A \\ &= \int_0^t F(A_s, X_s) dA_s. \end{aligned}$$

- Regarding the second sum a Taylor expansion w.r.t. the second variable yields

$$F(A_{t_i}, X_{t_{i+1}}) - F(A_{t_i}, X_{t_i}) = F_x(A_{t_i}, X_{t_i}) \Delta_i X + \frac{1}{2} F_{xx}(A_{t_i}, X_{t_i}) (\Delta_i X)^2 + R_{i,n}.$$

The remainder $R_{i,n}$ (in Lagrange form) is given by

$$R_{i,n} = \frac{1}{2} (F_{xx}(A_{t_i}, \tilde{x}_i) - F_{xx}(A_{t_i}, X_{t_i})) (\Delta_i X)^2 \quad \text{for some } \tilde{x}_i \in (X_{t_i}, X_{t_{i+1}}).$$

Corollary 3.2.4 applied to the continuous function $g(t) = F_{xx}(A_t, X_t)$ yields

$$\lim_{n \rightarrow \infty} \frac{1}{2} \sum_{t_i \in \Pi^n} F_{xx}(A_{t_i}, X_{t_i}) (\Delta_i X)^2 = \frac{1}{2} \int_0^t F_{xx}(A_s, X_s) d\langle X \rangle_s.$$

³We use the following convention: for a non-monotone function g by $a \in (g_{t_i}, g_{t_{i+1}})$ we mean that $a \in (g_{t_i}, g_{t_{i+1}})$ if $g_{t_i} \leq g_{t_{i+1}}$ and that $a \in (g_{t_{i+1}}, g_{t_i})$ else.

Thus, we have established convergence of $\sum_{t_i} F_x(A_{t_i}, X_{t_i}) \Delta_i X$ if we can show that

$$\lim_{n \rightarrow \infty} \sum_{t_i \in \Pi^n} R_{i,n} = 0.$$

To this end, we observe that X is uniformly continuous on $[0, T]$ so $\max_i |X_{t_{i+1}} - X_{t_i}| \leq \delta_n \rightarrow 0$. Hence the remainder term can be estimated from above by

$$|R_{i,n}| \leq \max\{|F_{xx}(A_{t_i}, x) - F_{xx}(A_{t_i}, y)| : |x - y| \leq \delta_n, t_i \in \Pi^n\} (\Delta_i X)^2.$$

Continuity of F_{xx} and A yields $\max\{|F_{xx}(A_t, x) - F_{xx}(A_t, y)| : |x - y| \leq \delta_n; t \in [0, T]\} \rightarrow 0$ as $n \rightarrow \infty$. Since $\sum (\Delta_i X) \rightarrow \langle X \rangle_T$ this shows that

$$\sum_i |R_{i,n}| \rightarrow 0.$$

□

A special case of the preceding formula is $A_t = t$. In this case the mapping f is of the form $f(t, X_t)$.

Corollary 3.2.7 *Let $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ be of class $C^{1,2}$ and $X : [0, T] \rightarrow \mathbb{R}$ be continuous with continuous quadratic variation $\langle X \rangle$. Then*

$$f(t, X_t) - f(0, X_0) = \int_0^t f_t(s, X_s) ds + \int_0^t f_x(s, X_s) dX_s + \frac{1}{2} \int_0^t f_{xx}(s, X_s) d\langle X \rangle_s.$$

Another immediate result is the product rule which is obtained by applying the Itô formula to the function $f(x, y) = xy$.

Corollary 3.2.8 (Product Rule) *Let (X, Y) be continuous with continuous quadratic variation. Then*

$$X_t Y_t - X_0 Y_0 = \int_0^t X_s dY_s + \int_0^t Y_s dX_s + \langle X, Y \rangle_t.$$

The Itô formula also allows us to evaluate the integral of smooth functions of BM w.r.t. BM.

Example 3.2.9 *Let B be a standard Brownian motion and $F : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable. Then*

$$F(B_t) = F(0) + \int_0^t F'(B_s) dB_s + \frac{1}{2} \int_0^t F''(B_s) ds. \quad (3.2)$$

If $F(x) = x^n$, then

$$X_t^n = X_0^n + n \int_0^t X_s^{n-1} dX_s + \frac{n(n-1)}{2} \int_0^t X_s^{n-2} d\langle X \rangle_s.$$

In particular, we obtain a closed form solution for the integral of a Brownian path with respect to itself:

$$B_t^2 = B_0^2 + 2 \int_0^t B_s dB_s + \int_0^t d\langle B \rangle_s = 2 \int_0^t B_s dB_s + t.$$

The following example is of particular interest in mathematical finance.

Example 3.2.10 (*Geometric Brownian Motion*) Given a BM B , and constants $\mu \in \mathbb{R}$ and $\sigma > 0$ the geometric Brownian motion is given by the process

$$S_t = S_0 \exp\left(\sigma B_t + \left(\mu - \frac{1}{2}\sigma^2\right)t\right).$$

In order to write this process in differential form we apply Itô's formula to the exponential function $x \mapsto e^x$ and to the process

$$X_t = \left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma B_t.$$

In view of Corollary [3.1.5](#) we have that $\langle X \rangle_t = \sigma^2 t$ so Itô's formula yields

$$\begin{aligned} S_t = S_0 \exp(X_t) &= S_0 + \int_0^t S_u dX_u + \frac{1}{2} \int_0^t S_u d\langle X \rangle_u \\ &= S_0 + \int_0^t S_u \left[\left(\mu - \frac{1}{2}\sigma^2\right) du + \sigma dB_u \right] + \frac{1}{2} \int_0^t \sigma^2 S_u du \\ &= S_0 + \int_0^t \mu S_u du + \int_0^t \sigma S_u dB_u. \end{aligned}$$

In differential form the geometric Brownian motion can be written as

$$\frac{dS_t}{S_t} = \mu dt + \sigma dB_t.$$

Thus, if we assume that a stock price grows at an average rate μ and that the growth rates fluctuate around their average rate according to σB_t , then the price dynamics follows a geometric Brownian motion. This is the assumption underlying the Black-Scholes option pricing model that we briefly consider in the next section.

3.2.2 The quadratic variation of Itô integrals

If $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuously differentiable and $G' = g$, then it follows from the Itô formula that the Itô integral

$$Y_t := \int_0^t g(X_s) dX_s$$

is well defined and it is easy to show that the mapping $t \mapsto Y_t$ is continuous. The latter uses the mean-value theorem of RS integrals. For many applications it will be important to identify the quadratic variation of Y . To this end, we first prove the following result.

Proposition 3.2.11 *Let X be real-valued, continuous and of continuous quadratic variation and $f \in C^1$. Then the mapping $t \mapsto f(X_t)$ is of continuous quadratic variation and*

$$\langle f(X) \rangle_t = \int_0^t (f'(X_s))^2 d\langle X \rangle_s.$$

PROOF: We apply the Taylor formula in order to obtain

$$f(X_{t_{i+1}}) - f(X_{t_i}) = f'(X_{t_i})\Delta_i X + (f'(\tilde{x}_i) - f'(X_{t_i}))\Delta_i X$$

for some $\tilde{x}_i \in (X_{t_i}, X_{t_{i+1}})$. Taking the squares and summing over all points in time yields

$$\sum_{t_i} (f(X_{t_{i+1}}) - f(X_{t_i}))^2 = \sum_{t_i} (f'(X_{t_i}))^2 (\Delta_i X)^2 + \sum_{t_i} R_{i,n}$$

where

$$R_{i,n} = 2f'(X_{t_i}) (f'(\tilde{x}_i) - f'(X_{t_i})) (\Delta_i X)^2 + (f'(\tilde{x}_i) - f'(X_{t_i}))^2 (\Delta_i X)^2.$$

Thus, continuity of f' and X guarantee that

$$\sum_{t_i} R_{i,n} \leq C_n \sum_{t_i} (\Delta_i X)^2$$

for some sequence $C_n \rightarrow 0$. Hence the assertion follows because $\sum_{t_i} (\Delta_i X)^2 \rightarrow \langle X \rangle_t$. $\square$

As a corollary to the preceding proposition we obtain a formula for the quadratic variation of Itô integrals.

Corollary 3.2.12 *Let X be real-valued, continuous and of continuous quadratic variation and $g \in C^1$ and $G' = g$. Then the process*

$$Y_t := \int_0^t g(X_s) dX_s$$

has quadratic variation

$$\langle Y \rangle_t = \int_0^t g^2(X_s) d\langle X \rangle_s.$$

PROOF: By Itô's formula

$$\int_0^t g(X_s) dX_s = G(X_t) - G(X_0) - \frac{1}{2} \int_0^t g'(X_s) d\langle X \rangle_s.$$

The integral on the r.h.s. is of bounded variation because the process $\langle X \rangle$ is increasing; specifically,

$$V_T \left(\int_0^\cdot g'(X_s) d\langle X \rangle_s, \Pi^n \right) \leq \|g'\|_\infty \langle X \rangle_T.$$

In particular, its quadratic variation is zero and so

$$\langle Y \rangle = \langle G(X) \rangle.$$

Hence, the assertion follows from the preceding proposition. $\square$

Example 3.2.13 *The quadratic variation process of the geometric Brownian motion*

$$\frac{dS_t}{S_t} = \mu dt + \sigma dB_t$$

is given by the process

$$t \mapsto \int_0^t \sigma^2 S_u^2 du.$$

The preceding example suggests that if we can observe a geometric Brownian motion with unknown drift μ and unknown volatility σ^2 continuously over a small time interval, then we can infer the unknown σ^2 from our observations. In order to make this more precise, let us assume that we are given the discrete observations S_{t_i} ($i = 0, \dots, n$) where we assume for simplicity that $t_{i+1} - t_i = \delta$. The logarithmic increments can then be represented as

$$R_i := \ln \frac{S_{t_i}}{S_{t_{i-1}}} = \left(\mu - \frac{1}{2} \sigma^2 \right) \delta + \sigma \sqrt{\delta} Z_i$$

where $Z_1, \dots, Z_n$ are independent standard normal random variables. In particular, $R_1, \dots, R_n$ are i.i.d. normal random variables with mean $(\mu - \frac{1}{2} \sigma^2) \delta$ and variance $\sigma^2 \delta$.

Definition 3.2.14 For a given family $X_1, \dots, X_n$ random variables with common density function $f(\cdot, \theta)$ with unknown parameter θ the likelihood function is given by the function

$$L(\theta) := f(X_1, \dots, X_n, \theta).$$

The maximizer $\hat{\theta}$ of the function $L(\theta)$ (assuming it is unique) is called the maximum likelihood estimator (MLE) of θ , given $X_1, \dots, X_n$. The MLE is called consistent if $\hat{\theta} \rightarrow \theta$ a.s. if $n \rightarrow \infty$.

Exercise 27 (i) Let $X_1, \dots, X_n$ independent normal random variables with unknown mean μ and unknown volatility σ^2 . Show that the MLE is consistent and given by

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i; \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2.$$

(ii) Let $S_{t_0}, \dots, S_{t_n}$ be the realizations of a geometric Brownian motion process with fixed sample frequency $\delta = t_i - t_{i-1}$ ($i = 1, \dots, n$) and unknown (μ, σ^2) . Show that

$$f(S_1, \dots, S_n, \mu, \sigma) = \left(\prod_{i=1}^n \frac{1}{S_i} \right) \frac{1}{\sigma^n (2\pi\delta)^{\frac{n}{2}}} \exp \left(-\frac{1}{2\sigma^2\delta} \sum_{i=1}^n (\ln S_{t_i} - \ln S_{t_{i-1}} - (\mu - \frac{1}{2}\sigma^2)\delta)^2 \right).$$

(iii) Show that the MLE for geometric Brownian motion is given by

$$\hat{\mu} = \frac{1}{2}\hat{\sigma}^2 + \frac{1}{\delta}\bar{R} \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{\delta} \frac{1}{n} \sum_{i=1}^n (R_i - \bar{R})^2$$

where $R_i := \ln \frac{S_{t_i}}{S_{t_{i-1}}}$ and $\bar{R} := \frac{1}{n} \sum_{i=1}^n R_i$.

In order to see if the estimators for the unknown parameters of the geometric Brownian motion obtained in the previous exercise are consistent, let us first notice that since $n\delta = t_n - t_0$,

$$\frac{1}{\delta}\bar{R} = \left(\mu - \frac{1}{2}\sigma^2 \right) + \sigma \frac{B_{t_n} - B_{t_0}}{t_n - t_0} \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{\delta} \frac{1}{n} \sum_{i=1}^n (R_i)^2 - \frac{1}{\delta} (\bar{R})^2.$$

If we hold the sampling frequency δ fix and increase the number of observation to infinity, then

$$\frac{1}{t_n - t_0} (B_{t_n} - B_{t_0}) = \frac{1}{n\delta} \sum_{i=1}^n (B_{t_i} - B_{t_{i-1}}) \rightarrow 0 \text{ a.s.}$$

by the independence of Brownian increments and the strong law of large numbers. In this case $\hat{\mu} \rightarrow \mu$ and $\hat{\sigma}^2 \rightarrow \sigma^2$ a.s. However, holding δ fix and increase the number of observations to infinity requires to observe the process S on an arbitrarily long time interval. If we only observe S on a finite time interval, say $[0, T]$ and increase instead the sampling frequency, i.e. let $\delta \rightarrow 0$, the situation is very different. In this case,

$$\frac{1}{\delta}\bar{R} = \left(\mu - \frac{1}{2}\sigma^2 \right) + \sigma \frac{B_T - B_{t_0}}{T - t_0},$$

which does not longer converge because now T is a fixed finite constant. Thus, the MLE for the drift is no longer consistent. But the MLE for the volatility still is. This shows that it is much more difficult to estimate the drift of a geometric BM from a single finite sample path than to estimate the volatility.

Exercise 28 Prove that $\hat{\sigma}^2 \rightarrow \sigma^2$ a.s.

3.3 Quadratic variation of continuous local martingales

So far, we proved that integrals of smooth functions of Brownian motion w.r.t. Brownian sample paths exist almost surely. The proof heavily relied on the path-wise existence of the quadratic variation of Brownian motion. In this section, we consider the quadratic variation of continuous local martingales of which BM is a particular example. We prove that the quadratic variation exists as a limit in probability. This will later allow us to analyse more general stochastic integrals.

3.3.1 Local martingales

We start with the definition a local martingale. Loosely speaking an adapted stochastic process is a local martingale if it is a martingale when suitably stopped.

Definition 3.3.1 *A stochastic process $M = (M_t)$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$ is called a local martingale if there exists a localising sequence of stopping times $\{\tau_n\}$, that is, a sequence of stopping times $\{\tau_n\}$ such that*

- $\tau_n \uparrow \infty$ a.s.
- the stopped process $M^{\tau_n} = M_{\tau_n \wedge \cdot}$ is a martingale w.r.t. $(\mathcal{F}_t)$ for every $n \in \mathbb{N}$.

The space of all continuous local martingales is denoted $\mathcal{M}_{loc}$; the space of continuous local martingales starting in zero is denoted $\mathcal{M}_{loc}^0$.

Localising sequences are not unique. For instance, if $\{\tau_n\}$ is a localising sequence, then so is $\{\tau_n \wedge n\}$. Obviously, any martingale is a local martingale: in this case we can take $\tau_n \equiv +\infty$ for all $n \in \mathbb{N}$. But not every local martingale is a (true) martingale; even uniform integrability is not enough. We give an example in Chapter 4.

Let us give two conditions that guarantee that a local martingale is a true martingale.

Lemma 3.3.2 *Let M be a local martingale such that the running maximum $M_t^* \in L^1$ for all $t \geq 0$. Then M is a martingale.⁴ In particular, every bounded local martingale is a martingale.*

PROOF: Let $\{\tau_n\}$ be a localising sequence and $s \leq t$. Then $\mathbb{E}[M_{t \wedge \tau_n} | \mathcal{F}_s] = M_{s \wedge \tau_n}$. Since $|M_{t \wedge \tau_n}| \leq |M_t^*|$ dominated convergence yields the desired result. $\square$

It turns out that continuous local martingales are martingales if and only if the stopped processes have certain uniform integrability properties. For this we define what it means for a local martingale to be of “class DL”.

Definition 3.3.3 *Let $\mathcal{T}$ be the class of all finite-valued stopping times and $\mathcal{T}_T$ be the class of all stopping times bounded by $T > 0$. A stochastic process X is said to be of class D, respectively of class DL if the set*

$$\{X_\tau : \tau \in \mathcal{T}\}, \quad \text{respectively,} \quad \{X_\tau : \tau \in \mathcal{T}_T\}$$

is uniformly integrable for all $T > 0$.

⁴Warning: The condition $\sup_t \mathbb{E}|M_t| < \infty$ is not sufficient for M to be a martingale.

Each uniformly integrable martingale is of class D, in particular of class DL. Conversely, if a local martingale M is of class DL, then it is a martingale and a uniformly integrable martingale if it is of class D. In fact, for any localising sequence $\{\tau_n\}$ we have that for any $s \leq t$ that

$$M_{s \wedge t_n} = \mathbb{E}[M_{t \wedge t_n} | \mathcal{F}_s].$$

Since M is of class DL, the sequences $\{M_{s \wedge t_n}\}_{n \in \mathbb{N}}$ and $\{M_{t \wedge t_n}\}_{n \in \mathbb{N}}$ are uniformly integrable. Hence, they converge to M_s respectively M_t both a.s. and in L^1 and so the limit passes through the above conditional expectation. This shows the martingale property. If the local martingale M is of class D, it is a-fortiori uniformly integrable and hence a uniformly integrable martingale.

We actually have the following slightly stronger result.

Proposition 3.3.4 *A continuous local martingale M is a martingale if and only if it is of class DL.*

PROOF: If M is a continuous martingale, then Doob's optional sampling theorem yields that

$$M_\tau = \mathbb{E}[M_T | \mathcal{F}_\tau] \quad \text{for all } \tau \in \mathcal{T}_T.$$

In particular $\{M_\tau : \tau \in \mathcal{T}_T\}$ is uniformly integrable. The converse has already been shown. $\square$

Exercise 29 *Let X be a non-negative local martingale with $\mathbb{E}X_0 < \infty$. Show that X is a supermartingale.*

Exercise 30 *Let X be a right continuous submartingale. Show that under any of the following conditions the set*

$$\{X_\tau : \tau \in \mathcal{T}_T\}$$

is uniformly integrable for all $T \geq 0$.

- $X_t \geq 0$ a.s. for all $t \geq 0$.
- X admits the Doob-Meyer decomposition $X_t = M_t + A_t$ where M is a martingale and A is an increasing adapted process.

We say that a stopping time τ reduces a process M if the stopped process M^τ is a uniformly integrable martingale.

Exercise 31 *Let M, N be local martingales and let σ, τ be stopping times. Show the following results*

- If τ reduces M and $\sigma \leq \tau$ a.s., then σ reduces M .
- The sum $M + N$ is a local martingale.
- If σ and τ both reduce M , then so does the $\sigma \vee \tau$.
- The processes M^τ and $M^\tau \mathbf{1}_{\{\tau > 0\}}$ are local martingales.
- Let X be a càdlàg process and let $\{\tau_n\}$ be a sequence of stopping times such that $\tau_n \uparrow \infty$ a.s. and such that $X^{\tau_n} \mathbf{1}_{\{\tau_n > 0\}}$ is a local martingale for each $n \in \mathbb{N}$. Then X is a local martingale.

Hint. Use the Optional Sampling Theorem.

Let X be a stochastic process. We say that a property holds locally if there exists a sequence of stopping times $\{\tau_n\}$ such that $\tau_n \uparrow \infty$ a.s. such that $X^{\tau_n} \mathbf{1}_{\{\tau_n > 0\}}$ has this property for all $n \in \mathbb{N}$. It follows from the preceding exercise that a process that is locally a local martingale is also a local martingale. The following exercise discusses other examples of local properties.

Exercise 32 • *Let X be a process that is locally a square integrable martingale. Then X is a local martingale.*

- *Let X be an adapted, càdlàg process and let $\{\tau_n\}$ be a sequence of stopping times such that $\tau_n \uparrow \infty$ a.s. If $X^{\tau_n} \mathbf{1}_{\{\tau_n > 0\}}$ is a martingale for each $n \in \mathbb{N}$, then M is a local martingale.*

3.3.2 The quadratic variation process

If M is a local martingale, then M^2 is a local sub-martingale, and hence it has increasing expected values. The next theorem states that we can compensate for the increase in expected values by subtracting the quadratic variation of the trajectories of M , which exist as limits in probability. In the case of Brownian motion they exist as almost sure limits.

Theorem 3.3.5 *Let M be a continuous local square integrable martingale with respect to a filtration that satisfies the usual conditions. Let $M_0 = 0$. Then there exists an almost surely unique continuous process $\langle M \rangle$ (called the bracket process, compensator or quadratic variation process) such that*

- $\langle M \rangle_0 = 0$ and $\langle M \rangle$ is non-decreasing
- $M^2 - \langle M \rangle$ is a continuous local martingale
- for any sequence of partitions $\{\Pi^n\}$ of $[0, t]$ with $|\Pi^n| \rightarrow 0$,

$$V_t^2(M(\omega), \Pi^n) \rightarrow \langle M \rangle_t \text{ uniformly on compact time intervals in probability (ucp)}$$

The proof is given below. It follows from the proof that if M is a square integrable continuous martingale, then $M^2 - \langle M \rangle$ is a continuous martingale and

$$\mathbb{E}M_t^2 = \mathbb{E}\langle M \rangle_t, \quad t \geq 0.$$

Using Theorem 3.3.5 along with the (uniform) continuity of M (on compact time intervals) yields the following result.

Corollary 3.3.6 *Let A be a process of bounded variation, and let M be a continuous local martingale starting in zero. Then*

$$V_t^2(M(\omega) + A(\omega), \Pi^n) \rightarrow \langle M \rangle_t \text{ ucp.}$$

Thus, we can write

$$\langle M + A \rangle = \langle M \rangle.$$

We also have the following result on the bracket process of stopped local martingales.

Corollary 3.3.7 *Let M be a continuous square integrable local martingale and τ be a stopping time. Then*

$$\langle M^\tau \rangle_t = \langle M \rangle_{\tau \wedge t}.$$

PROOF: By the optional sampling theorem, the process $t \mapsto M_{\tau \wedge t}^2 - \langle M \rangle_{\tau \wedge t}$ is a local martingale. Hence the assertion follows by the uniqueness of the bracket process. $\square$

The following exercise encourages you to review several important results on uniform integrable martingales and their long run behavior that have been established in Section 1.

Exercise 33 *Let M be a continuous, square integrable local martingale and assume that $\mathbb{E}\langle M \rangle_\infty < \infty$. Then the following holds.*

- i) M is a martingale and both M and M^2 are uniformly integrable.
- ii) The limit M_∞ exists a.s. and $\mathbb{E}M_\infty^2 = \mathbb{E}\langle M \rangle_\infty$.
- iii) We may take the right-continuous modification of

$$Z_t := \mathbb{E}[M_\infty^2 | \mathcal{F}_t] - M_t^2$$

which is a right-continuous (by appropriate choice of the modification), non-negative super-martingale and satisfies $\mathbb{E}[Z_t] \rightarrow 0$ for $t \rightarrow \infty$

Hint. You may use without proof that for any continuous local martingale M and any stopping time τ it holds that $\mathbb{E}[M_\tau^2] \leq \mathbb{E}\langle M \rangle_\tau$ where $M_\infty^2 := \liminf_{t \rightarrow \infty} M_t^2$. Also, recall that an L^2 -bounded family of random variables is uniformly integrable.

The covariation process and orthogonality of martinagles

For (right-)continuous, local square integrable martingales M and N we define their crossvariation $\langle M, N \rangle$ through the polarisation identity:

$$\langle M, N \rangle := \frac{1}{4} (\langle M + N \rangle - \langle M - N \rangle). \quad (3.3)$$

Clearly, the $(M, N) \mapsto \langle M, N \rangle$ is bilinear and symmetric. It is not difficult to show following result. The proof is left as an exercise.

Proposition 3.3.8 *Let M, N be continuous, square integrable local martingales. Let $\{\Pi^n\}$ be any sequence of partitions of $[0, t]$ with $|\Pi^n| \rightarrow 0$. Then*

$$\sum_{i=1}^m (M_{t_{i+1}^n} - M_{t_i^n})(N_{t_{i+1}^n} - N_{t_i^n}) \rightarrow \langle M, N \rangle_t \quad \text{ucp.}$$

Remark 3.3.9 *For $0 \leq s < t$ the following defines a defines a symmetric positiv definite bilinear form as well:*

$$\langle M, N \rangle_s^t := \langle M, N \rangle_t - \langle M, N \rangle_s.$$

Hence it follows from the usual Cauchy-Schwarz inequality that a.s.

$$|\langle M, N \rangle_t - \langle M, N \rangle_s| \leq \sqrt{\langle M \rangle_t - \langle M \rangle_s} \sqrt{\langle N \rangle_t - \langle N \rangle_s}.$$

This implies that

$$\mathbb{E} \left[\sup_{0 \leq s \leq t} |\langle M, N \rangle_s| \right] \leq \left(\mathbb{E} \sqrt{\langle M \rangle_t} \right) \left(\mathbb{E} \sqrt{\langle N \rangle_t} \right). \quad (3.4)$$

It is easy to see that

$$MN - \langle M, N \rangle = \frac{(M + N)^2 - \langle M + N \rangle}{4} - \frac{(M - N)^2 - \langle M - N \rangle}{4},$$

which is the difference of two continuous local martingales and hence a continuous local martingale. We call M, N orthogonal if $\langle M, N \rangle = 0$. Since for any $t \geq s$,

$$\mathbb{E}[(M_t - M_s)(N_t - N_s) | \mathcal{F}_s] = \mathbb{E}[M_t N_t - M_s N_s | \mathcal{F}_s] = \mathbb{E}[\langle M, N \rangle_t - \langle M, N \rangle_s | \mathcal{F}_s]$$

we see that orthogonality of square integrable martingales means their product is martingale and their increments are conditionally uncorrelated.

The converse is also true as shown by the following lemma. A continuous local martingale w.r.t. a filtration that satisfies the usual conditions is constant if it is of bounded variation. Hence, if the filtration satisfies the usual conditions, then the above equality shows that orthogonality of M, N is equivalent to the product MN being a martingale.

Lemma 3.3.10 *Let M be a continuous local martingale with respect to a filtration that satisfies the usual conditions starting in 0. If M is of bounded variation (that is, $V_t(M(\omega)) < \infty$ a.s. for all $t \geq 0$), then $\mathbb{P}[M_t = 0 \text{ for all } t \geq 0] = 1$.*

PROOF: By the continuity of the sample paths, it is enough to prove that $\mathbb{P}[M_t = 0] = 1$ for all $t \geq 0$. To this end, we proceed in three steps.

Let us first assume that M is a continuous square integrable martingale and satisfies $V_t(M(\omega)) \leq C$ for almost all $\omega \in \Omega$. Then, for any partition $0 = t_1 < \dots < t_n = t$,

$$\mathbb{E}M_t^2 = \mathbb{E} \left[\sum_i \Delta_i M \right]^2 = \sum_i \mathbb{E}[\Delta_i M]^2 \leq C \mathbb{E} \left[\sup_i |\Delta_i M| \right]$$

By continuity, $\sup_i |\Delta_i M| \rightarrow 0$ a.s. if $|\Pi^n| \rightarrow 0$. Moreover, by Doob's maximal inequality

$$\mathbb{E} \left[\sup_{n \in \mathbb{N}, t_i \in \Pi^n, t_i \leq t} |\Delta_i M|^2 \right] \leq 2 \mathbb{E} \left[\sup_{0 \leq u \leq t} |M_u|^2 \right] \leq C \mathbb{E}|M_t|^2 < \infty.$$

In particular, $\sup_{n \in \mathbb{N}, t_i \in \Pi^n} |\Delta_i M| \in L^2$. Hence, by dominated convergence $\mathbb{E}M_t^2 = 0$ and $\mathbb{P}(M_t = 0) = 1$.

Let us now assume that M is a continuous square integrable martingale of bounded (yet not uniformly bounded) variation and put

$$\tau^n := \inf\{t \geq 0 : V_t(M(\omega)) > n\}.$$

Then

$$\{t \leq \tau^n\} = \{V_t(M(\omega)) \leq n\} \in \mathcal{F}_t.$$

Thus τ^n is an optional time and hence a stopping time because the filtration satisfies the usual conditions. By optional sampling, the stopped process M^{τ^n} is a martingale and by the first step $\mathbb{E}M_{\tau^n \wedge t}^2 = 0$. Using that $\tau^n \uparrow \infty$ by assumption we obtain from $\mathbb{E}|M_t^*|^2 < \infty$ that $\mathbb{E}|M_t|^2 = 0$.

Let us finally assume that M is a continuous local martingale with localising sequence $\{\tau^n\}$. We may w.l.o.g. assume that M^{τ^n} is a bounded martingale; else we can define

$$\sigma^n := \tau^n \wedge \inf\{t \geq 0 : |M_t| \geq n\},$$

which is again a localising sequence. Hence $\mathbb{P}(M_{t \wedge \tau^n} = 0) = 1$. Since $\tau^n \uparrow \infty$ a.s. dominated convergence yields $\mathbb{P}(M_t = 0) = 1$. $\square$

Exercise 34 Let M be a continuous local martingale with continuous quadratic variation and τ be a stopping time. If $\langle M \rangle_\tau = 0$, then $\mathbb{P}[X_{\tau \wedge t} = 0, t \in [0, \infty)] = 1$.

Some useful martingale properties

In what follows we will repeatedly use the following property of a square integrable martingale M : for every $0 \leq s \leq t \leq u < v < \infty$,

$$\begin{aligned} \mathbb{E}[(M_v - M_u)^2 | \mathcal{F}_t] &= \mathbb{E}[M_v^2 - M_u^2 | \mathcal{F}_t] \\ \mathbb{E}[(M_v - M_u)(M_t - M_s)] &= 0. \end{aligned} \tag{3.5}$$

This means that when squaring the sums of martingale increments and taking the expectation, we can neglect the cross-product terms. In particular,

$$\mathbb{E}[(M_v - M_u)^2 | \mathcal{F}_t] = \mathbb{E}[\langle M \rangle_v - \langle M \rangle_u | \mathcal{F}_t].$$

Thus, the terms $M_v^2 - \langle M \rangle_v - (M_u^2 - \langle M \rangle_u)$ and $(M_v^2 - M_u^2) - (\langle M \rangle_v - \langle M \rangle_u)$ have the same conditional expectation given $\mathcal{F}_t$, namely zero. In particular, the expectation of products of such terms over nonoverlapping intervals is still zero.

In what follows, for a given partition $\Pi = \{t_0 < t_1 < \dots < t_m\}$ we put

$$\Delta_i M := M_{t_i} - M_{t_{i-1}}, \text{ for } i = 1, \dots, m$$

3.3.3 Proof of Theorem 3.3.5 for bounded martingales

In this section we prove Theorem 3.3.5 under the additional assumption that M is a continuous martingale (not just a local martingale) and uniformly bounded; that is we assume in this section that $|M_t| \leq K$, for some $K > 0$. We start with the following auxiliary result.

Lemma 3.3.11 Let M be a martingale bounded by K with $M_0 = 0$. For any partition $\Pi = \{0 = t_0 < t_1 < \dots < t_m = t\}$ of $[0, t]$,

$$\mathbb{E}[V_t^2(M, \Pi)]^2 \leq 6K^4.$$

PROOF: We have that

$$\mathbb{E}[V_t^2(M, \Pi)]^2 = \mathbb{E}\left[\sum_{i=1}^m (\Delta_i M)^4\right] + 2\mathbb{E}\left[\sum_{k=1}^{m-1} \sum_{j=k+1}^m (\Delta_j M)^2 (\Delta_k M)^2\right]$$

Using (3.5) (resulting in a telescope sum) and the fact that $M_0 = 0$ we get

$$\mathbb{E}\left[\sum_{i=1}^m (\Delta_i M)^4\right] \leq 4K^2 \mathbb{E}\left[\sum_{i=1}^m (\Delta_i M)^2\right] \leq 4K^2 \mathbb{E}M_t^2 \leq 4K^4.$$

Furthermore, for $k \leq m-1$,

$$\mathbb{E}\left[\sum_{j=k+1}^m (\Delta_j M)^2 | \mathcal{F}_{t_k}\right] = \mathbb{E}\left[\sum_{j=k+1}^m (M_{t_j}^2 - M_{t_{j-1}}^2) | \mathcal{F}_{t_k}\right] \leq K^2.$$

Hence, we obtain for the second term that

$$\begin{aligned} 2\mathbb{E}\left[\sum_{k=1}^{m-1}\sum_{j=k+1}^m(\Delta_j M)^2(\Delta_k M)^2\right] &= 2\mathbb{E}\left[\sum_{k=1}^{m-1}(\Delta_k M)^2\mathbb{E}\left[\sum_{j=k+1}^m(\Delta_j M)^2|\mathcal{F}_{t_k}\right]\right] \\ &\leq 2K^2\mathbb{E}\left[\sum_{k=1}^{m-1}(\Delta_k M)^2\right] \leq 2K^4. \end{aligned}$$

□

Corollary 3.3.12 *Let M be a bounded continuous martingale starting in zero, and let $V_t^4(M, \Pi)$ denote the sum of the fourth moments of the increments along the partition Π of $[0, t]$. Then,*

$$\lim_{|\Pi^n| \rightarrow 0} \mathbb{E}V_t^4(M, \Pi^n) = 0.$$

PROOF: For a given partition Π let

$$m_t(M, \Pi) := \sup\{|M_r - M_s| : 0 \leq r < s \leq t, |s - t| \leq |\Pi|\}.$$

Since M is uniformly continuous on $[0, t]$, we get $\lim_{|\Pi^n| \rightarrow 0} m_t(M, \Pi^n) = 0$ a.s. For any partition Π^n ,

$$V_t^4(M, \Pi^n) \leq m_t^2(M, \Pi^n)V_t^2(M, \Pi^n).$$

By Hölder's inequality,

$$\mathbb{E}V_t^4(M, \Pi^n) \leq \sqrt{\mathbb{E}[m_t^4(M, \Pi^n)]}\sqrt{\mathbb{E}[V_t^2(M, \Pi^n)]^2}.$$

Thus, the result follows from the fact that $V_t^2(M, \Pi^n)$ is L^2 bounded (by the preceding lemma). □

Construction of the bracket process

For bounded, continuous martingales we are now going to construct the process $\langle M \rangle$ as an L^2 limit (in the set of set of continuous functions) of quadratic increments along an *increasing* sequence of partitions $\{\Pi^n\}$, $\Pi^n = \{t_0^n < t_1^n < \dots < t_{m_n}^n\}$. Specifically, let

$$A_t^n := \sum_{t_i^n \in \Pi^n, t_i^n \leq t} (\Delta_i M)^2 + (M_t - M_{t_{m_n}^n})^2.$$

Using our general observation on the expectations of martingale increments and setting $t := t_{m_n+1}^n$ we see that the sum of products of martingale increments

$$N_t^n := M_t^2 - A_t^n = 2 \sum_{i=1}^{m_n+1} M_{t_{i-1}^n} \Delta_i M$$

is a *continuous* martingale. Hence, for any two partitions $\Pi^n \subseteq \Pi^m$, the difference $(N^n - N^m)$ is a continuous martingale and it follows from Doob's maximal inequality that

$$\mathbb{E}\left[\sup_{0 \leq s \leq t} |N_s^n - N_s^m|^2\right] \leq C\mathbb{E}[|N_t^n - N_t^m|^2]. \quad (3.6)$$

Since $\Pi^n \subseteq \Pi^m$ we can write

$$\dots < t_{i-1}^n = t_{j_{n_i}}^m < t_{j_{n_i+1}}^m < \dots < t_{j_{n_{i+1}}}^m = t_i^n < \dots$$

Moreover, putting $\Delta_j^n M = M_{t_j^n} - M_{t_{j-1}^n}$ the variable N_t^n can be expressed as

$$N_t^n = 2 \sum_i \sum_{j=n_i+1}^{n_{i+1}} M_{t_{j-1}^n} \Delta_j^n M$$

while N_t^m can be expressed as

$$N_t^m = 2 \sum_i \sum_{j=n_i+1}^{n_{i+1}} M_{t_{j-1}^m} \Delta_j^m M.$$

As a result, (assuming w.l.o.g. that t belongs to both Π^n and Π^m)

$$N_t^n - N_t^m = 2 \sum_i \sum_{j=n_i+1}^{n_{i+1}} (M_{t_{i-1}^n} - M_{t_{j-1}^m}) \Delta_j^m M.$$

This is a sum of martingale increments over non-overlapping intervals. Hence the expectation of the square of the sum equals the sum of the square terms, and an application of the Cauchy-Schwarz inequality yields

$$\begin{aligned} \mathbb{E}[|N_t^n - N_t^m|^2] &= 2 \mathbb{E} \left[\sum_i \sum_{j=n_i+1}^{n_{i+1}} (M_{t_{i-1}^n} - M_{t_{j-1}^m})^2 (\Delta_j^m M)^2 \right] \\ &\leq 2 \mathbb{E} \left[\sup_{i \in \{1, \dots, m_n\}, j \in \{n_i+1, \dots, n_{i+1}\}} |M_{t_{i-1}^n} - M_{t_{j-1}^m}|^2 \sum_j (\Delta_j^m M)^2 \right] \\ &\leq 2 \sqrt{\mathbb{E} \left[\sup_{i \in \{1, \dots, m_n\}, j \in \{n_i+1, \dots, n_{i+1}\}} |M_{t_{i-1}^n} - M_{t_{j-1}^m}|^4 \right]} \sqrt{\mathbb{E} \left[\sum_j (\Delta_j^m M)^2 \right]^2}. \end{aligned}$$

Boundedness along with continuity of M implies that the first term tends to zero if $n, m \rightarrow \infty$. The second term is bounded, due to Lemma 3.3.11. In view of (3.6) this proves that

$$\lim_{m, n \rightarrow \infty} \mathbb{E} \left[\sup_{0 \leq s \leq t} |N_s^n - N_s^m|^2 \right] = 0$$

and hence that $\{N^m\}$ forms a Cauchy sequence in $L^2(\Omega, C([0, T], \mathbb{R}))$ for all $T \geq 0$, where $C([0, T], \mathbb{R})$ is equipped with the supremum norm. Hence there exists $N \in L^2(\Omega, C([0, T], \mathbb{R}))$ such that $N^m \rightarrow N$ in $L^2(\Omega, C([0, T], \mathbb{R}))$. The martingale property of N will follow from more general results proved in below; see the proof of Theorem 4.2.2.

If we now define

$$\langle M \rangle_t := M_t^2 - N_t,$$

then $M^2 - \langle M \rangle$ is a continuous martingale and $A_t^n \rightarrow \langle M \rangle_t$ in L^2 uniformly on compact time intervals. This proves the existence of the bracket process.

Uniqueness of the bracket process

To see the uniqueness of the bracket process, let us assume that there exist two continuous, non-decreasing processes A, A' starting in zero such that $M^2 - A$ and $M^2 - A'$ are continuous local martingales. Then $A - A'$ is a continuous local martingale of bounded variation (as the difference of two non-decreasing processes) starting in zero. Hence it follows from Lemma 3.3.10 that $A = A'$ a.s.

3.3.4 Proof of Theorem 3.3.5

Let us now consider the general case of a continuous local martingale M that is not uniformly bounded. Let $\{\tau^n\}$ be a localising sequence and put

$$M_t^n := M_{\tau^n \wedge t}.$$

We can w.l.o.g. assume that M^n is bounded; else we can consider the localising sequence

$$\sigma^n := \tau^n \wedge \inf\{t \geq 0 : |M_t| \geq n\},$$

Let $\langle M^n \rangle$ be the process constructed in the previous section. By construction and the uniqueness of the bracket process we have for all $n \geq m$ that

$$\langle M^n \rangle \mathbf{1}_{[0, \tau^m]} = \langle M^m \rangle \mathbf{1}_{[0, \tau^m]}.$$

The sequence $\langle M^n \rangle_t$ is increasing in $n \in \mathbb{N}$ because $\tau^n \uparrow$. Hence, we can define the limit

$$\langle M \rangle_t := \lim_{n \rightarrow \infty} \langle M^n \rangle_t.$$

The limit is continuous as a pointwise limit of continuous functions, and finite on $[0, \tau^m]$ for all $m \in \mathbb{N}$ because $\langle M \rangle \mathbf{1}_{[0, \tau^m]} = \langle M^m \rangle \mathbf{1}_{[0, \tau^m]}$. Moreover,

$$(M^2 - \langle M \rangle)_{\tau^m \wedge \cdot} = ((M^m)^2 - \langle M^m \rangle)_{\tau^m \wedge \cdot}.$$

is a martingale because the term on the right-hand side is a martingale without stopping and hence a martingale after stopping by the optional sampling theorem. This proves that $M^2 - \langle M \rangle$ is a local martingale. The facts that $\langle M \rangle_0 = 0$ and that $\langle M \rangle$ is non-decreasing follow from construction. Uniqueness of such a process follows again from Lemma 3.3.10.

To prove the convergence in probability of $V_t^2(M, \Pi^m)$ to $\langle M \rangle_t$ for any $t \geq 0$ we recall that the following holds:

•

$$V_t^2(M^n, \Pi^m) \rightarrow \langle M^n \rangle_t \quad \text{in } L^2 \text{ and hence in probability as } m \rightarrow \infty$$

•

$$M^n \equiv M \text{ on } [0, \tau^n] \quad \text{and hence} \quad V^2(M^n, \Pi^m) \equiv V^2(M, \Pi^m) \text{ on } [0, \tau^n] \text{ for all } m \in \mathbb{N}.$$

Since $\{\tau^n\}$ is a localizing sequence, we know that

$$\lim_{n \rightarrow \infty} \mathbb{P}[\tau^n < t] = 0.$$

In particular, for every $\varepsilon > 0$ there exists $n(\varepsilon) \in \mathbb{N}$ such that⁵

$$\mathbb{P}[\tau^n \leq t] \leq \varepsilon \quad \text{for all } n \geq n(\varepsilon).$$

⁵This is the reason why we can only expect convergence in probability.

Hence, for $n \geq n(\varepsilon)$

$$\mathbb{P}[\langle M \rangle_t = \langle M^n \rangle_t \text{ on } [0, t]] \geq 1 - \varepsilon$$

and

$$\mathbb{P}[V^2(M^n, \Pi^m) \equiv V^2(M, \Pi^m) \text{ on } [0, t] \text{ for all } m \in \mathbb{N}] \geq 1 - \varepsilon.$$

As a result, there exists $m_0 = m(\varepsilon/3, n(\varepsilon/3)) \in \mathbb{N}$ such that for all $m \geq m_0$ the set

$$\begin{aligned} \left\{ \sup_{0 \leq t \leq T} |V_t^2(M, \Pi^m) - \langle M \rangle_t| < \varepsilon \right\} \supseteq & \left\{ \sup_{0 \leq t \leq T} |V_t^2(M, \Pi^m) - V_t^2(M^{n(\varepsilon/3)}, \Pi^m)| < \varepsilon/3 \right\} \\ & \cap \left\{ \sup_{0 \leq t \leq T} |V_t^2(M^{n(\varepsilon/3)}, \Pi^m) - \langle M^{n(\varepsilon/3)} \rangle_t| < \varepsilon/3 \right\} \\ & \cap \left\{ \sup_{0 \leq t \leq T} |\langle M^{n(\varepsilon/3)} \rangle_t - \langle M \rangle_t| < \varepsilon/3 \right\} \end{aligned}$$

has measure at least $1 - \varepsilon$. This proves the desired convergence in probability of the quadratic variation to the bracket process $\langle M \rangle$. $\square$

Remark 3.3.13 Notice that the process $\langle M \rangle$ can be defined as an L^2 -limit for bounded martingales but only as a limit in probability for unbounded, respectively local martingales.

